PROJECT ROSTER — Trustworthy Chest X-ray Deep L

Everyone owns one DNN model end-to-end (train → calibrate → explain → robu

Code	Member / Role	DNN Model Owned
M1	Member 1 — ResNet50 & Reproducibility/Lit Lead	ResNet50 (CNN)
M2	Member 2 — DenseNet121 & Data Pipeline Lead	DenseNet121 (CNN)
M3	Member 3 — EfficientNet-B0 & Efficiency Lead	EfficientNet-B0 (CNN)
M4	Member 4 — ViT-Base & Explainability Lead	ViT-Base (Transformer)
M5	Member 5 — Segmentation/Eval & Paper Lead	U-Net lung segmentation

LEGEND

Status va	Not started / In progress / Blocked / Done (dropdown in Master Ta
Milestone	M0 End W1 · M1 End W2 · M2 Mid W3 · M3 End W3 · M4 End W4 (se
[GATE]	A checkpoint that blocks downstream work until signed off.
Owner	Person accountable (does the work). Support = assists / reviews (RAC
Week	W1–W4 of the one-month plan.
Trust axe	Accuracy · Calibration · Explainability/Faithfulness · Robustness (OOD

earning (5-member team)

stness → efficiency) AND one shared lead role AND paper sections.

Shared Lead Role	Paper Sections
Repo, experiment-tracking, seeds, references	Intro, Related Work, Gap, Conclusion, Abstract
Preprocessing, augmentation, split, dataloaders (shared)	Methodology
Efficiency harness, Pareto, compute setup	Slides
XAI standard (Grad-CAM/rollout/SHAP/faithfulness), figures	Discussion
Calibration + robustness protocol, external data, stats	Results, final assembly

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Key Deliverables
ResNet50 full pipeline; reproducible repo; literature
DenseNet121 full pipeline; shared data pipeline; baseline
EfficientNet full pipeline; efficiency benchmark; Pareto
ViT full pipeline; XAI module; all figures
Segmentation; calibration+OOD protocol; consolidation; paper

MAIN MILESTONES & GATES

Milestone	Due	Name
M0	End of Week 1	Foundations Ready
M1	End of Week 2	All Models Trained
M2	Mid Week 3	Trust Experiments Done
M3	End of Week 3	Robustness & Efficiency Done
M4	End of Week 4	Paper & Slides Camera-Ready

Gate criteria (must all be true to pass)	Sign-off owner
Shared data pipeline frozen & reproducible; single baseline (DenseNet121) trains end-to-end; external data prepared; lit-review drafted.	M2 / M1
All 4 classification models trained (frozen + fine-tuned) with accuracy tables; U-Net segmentation model ready.	Model owners / M5
Calibration + explainability + faithfulness completed for all 4 models under shared standards.	M4 / M5
External OOD test + efficiency benchmark for all models; segmentation-ablation (shortcut) experiment complete.	M5 / M3
Consolidated analysis, Pareto trade-off, full paper draft, slides; reproducibility rerun passes.	M1 / M5

MASTER TASK LIST — single source of truth (update Status in column J)

ID	Epic	Task / Sub-issue	Owner	Support
T01	0 Setup	Create GitHub repo, branch strategy & folder layout (data/ src/ notebooks/ results/ paper/)	**M1**	All
T02	0 Setup	Set up experiment tracking (W&B/TensorBoard) + fixed-seed & reproducibility standard	**M1**	M5
T03	0 Setup	Compute setup: Colab/Kaggle GPU, requirements.txt, env pinning	**M3**	M2
T04	0 Setup	Define shared config: image size, split ratios, metrics schema, checkpoint format	**M5**	All
T05	1 Data	Download COVID-19 Radiography dataset; integrity check; class-count report	**M2**	M1
T06	1 Data	Build preprocessing module (resize 224, normalise, grayscale->3ch)	**M2**	M3
T07	1 Data	Stratified 70/15/15 split with fixed seed; save index files	**M2**	M5
T08	1 Data	Augmentation module (rotation, flip, jitter, zoom, shift, crop)	**M2**	M4
T09	1 Data	Shared PyTorch DataLoader with class weights for imbalance	**M2**	M1
T10	1 Data	Download external RSNA set, DICOM->PNG convert, map labels	**M5**	M2
T11	2 Baseline	**Train single baseline (DenseNet121) end-to-end to validate pipeline [GATE M0]**	**M2**	All
T20	2 Train	Standardise shared training-loop template (frozen->fine-tune)	**M5**	M2
T12	2 Train	ResNet50 — Phase 1 frozen training	**M1**	-
T13	2 Train	ResNet50 — Phase 2 fine-tune + hyperparameter tuning	**M1**	-
T14	2 Train	DenseNet121 — Phase 1 frozen training	**M2**	-
T15	2 Train	DenseNet121 — Phase 2 fine-tune + tuning	**M2**	-
T16	2 Train	EfficientNet-B0 — Phase 1 frozen training	**M3**	-
T17	2 Train	EfficientNet-B0 — Phase 2 fine-tune + tuning	**M3**	-
T18	2 Train	ViT-Base — Phase 1 frozen training	**M4**	-
T19	2 Train	ViT-Base — Phase 2 fine-tune (DeiT/ViT-S fallback if unstable)	**M4**	-
T21	3 Segment	Train/adapt U-Net lung-segmentation model	**M5**	M2

T22	3 Segment	Generate lung masks for primary + external sets	**M5**	M2
T23	4 Calib	**Calibration standard module: ECE, Brier, reliability, temperature scaling [GATE E5]**	**M5**	M2
T24	4 Calib	ResNet50 calibration (pre/post temp scaling)	**M1**	M5
T25	4 Calib	DenseNet121 calibration	**M2**	M5
T26	4 Calib	EfficientNet-B0 calibration	**M3**	M5
T27	4 Calib	ViT-Base calibration	**M4**	M5
T28	5 XAI	**XAI standard: Grad-CAM (CNN), Attention Rollout (ViT), SHAP + faithfulness/localisation score [GATE E6]**	**M4**	M3
T29	5 XAI	ResNet50 explainability + faithfulness score	**M1**	M4
T30	5 XAI	DenseNet121 explainability + faithfulness	**M2**	M4
T31	5 XAI	EfficientNet-B0 explainability + faithfulness	**M3**	M4
T32	5 XAI	ViT-Base attention rollout + faithfulness	**M4**	-
T33	6 Robust	External-test protocol standard (label mapping, OOD metrics)	**M5**	M1
T34	6 Robust	ResNet50 external test + explanation-shift analysis	**M1**	M5
T35	6 Robust	DenseNet121 external test	**M2**	M5
T36	6 Robust	EfficientNet-B0 external test	**M3**	M5
T37	6 Robust	ViT-Base external test	**M4**	M5
T38	6 Robust	Full-image vs lung-segmented experiment (shortcut mitigation)	**M5**	M2
T39	7 Effic	**Efficiency harness: params, FLOPs, latency (CPU/GPU), GPU memory [GATE E8]**	**M3**	M5
T40	7 Effic	Run efficiency benchmark on all 4 models	**M3**	M5
T41	8 Analysis	Consolidate all metrics into master results tables	**M5**	M1
T42	8 Analysis	Statistical comparison (significance, confidence intervals)	**M5**	M1
T43	8 Analysis	Pareto trade-off analysis (accuracy vs trust vs efficiency)	**M3**	M5
T44	8 Analysis	Generate all figures (curves, heatmaps, reliability, Pareto)	**M4**	M3

T45	9 Paper	Introduction + Related Work + Research Gap	**M1**	M5
T46	9 Paper	Methodology section	**M2**	M1
T47	9 Paper	Experiments & Results section	**M5**	All
T48	9 Paper	Discussion + model-selection guidance	**M4**	M5
T49	9 Paper	Conclusion + Abstract	**M1**	M5
T50	9 Paper	Presentation slides	**M3**	All
T51	9 Paper	Reference management (Zotero) — ongoing	**M1**	All
T52	9 Paper	**Final proofread + reproducibility rerun from clean repo [GATE M4]**	**M5**	All

Week	M.stone	Depends on	Est. days	Status	Notes
W1	M0	-	1	Not started	
W1	M0	T01	1	Not started	
W1	M0	T01	1	Not started	
W1	M0	T01	1	Not started	
W1	M0	-	1	Not started	
W1	M0	T04	1	Not started	
W1	M0	T06	1	Not started	
W1	M0	T06	1	Not started	
W1	M0	T07,T08	1	Not started	
W1-W2	M1	T06	2	Not started	
W1	M0	T09	2	Not started	
W1-W2	M0	T09	1	Not started	
W2	M1	T09,T20	1	Not started	
W2	M1	T12	2	Not started	
W2	M1	T09,T20	1	Not started	
W2	M1	T14	2	Not started	
W2	M1	T09,T20	1	Not started	
W2	M1	T16	2	Not started	
W2	M1	T09,T20	1	Not started	
W2	M1	T18	2	Not started	
W2	M1	T06	2	Not started	

W2-W3	M1	T21	1	Not started	
W2-W3	M2	-	1	Not started	
W3	M2	T23,T13	1	Not started	
W3	M2	T23,T15	1	Not started	
W3	M2	T23,T17	1	Not started	
W3	M2	T23,T19	1	Not started	
W2-W3	M2	T22	2	Not started	
W3	M2	T28	1	Not started	
W3	M2	T28	1	Not started	
W3	M2	T28	1	Not started	
W3	M2	T28	1	Not started	
W3	M3	T10	1	Not started	
W3	M3	T33,T13	1	Not started	
W3	M3	T33,T15	1	Not started	
W3	M3	T33,T17	1	Not started	
W3	M3	T33,T19	1	Not started	
W3	M3	T22	2	Not started	
W3	M3	-	1	Not started	
W3	M3	T39,T13,T15,T17,T19	1	Not started	
W4	M4	T24,T29,T34,T40	1	Not started	
W4	M4	T41	1	Not started	
W4	M4	T41	1	Not started	
W4	M4	T41	1	Not started	

W1-W4	M4	-	3	Not started	
W2-W4	M4	-	2	Not started	
W4	M4	T41	2	Not started	
W4	M4	T41	2	Not started	
W4	M4	T47	1	Not started	
W4	M4	T41	2	Not started	
W1-W4	M4	-	1	Not started	
W4	M4	T47,T48,T49	1	Not started	

Member 1 — ResNet50 & Reproducibility/Lit Lead

TASKS YOU OWN (accountable)

ID	Epic	Task / Sub-issue	Week	M.stone
T01	0 Setup	Create GitHub repo, branch strategy & folder layout (data/ src/ notebooks/ results/ paper/)	W1	M0
T02	0 Setup	Set up experiment tracking (W&B/TensorBoard) + fixed-seed & reproducibility standard	W1	M0
T12	2 Train	ResNet50 — Phase 1 frozen training	W2	M1
T13	2 Train	ResNet50 — Phase 2 fine-tune + hyperparameter tuning	W2	M1
T24	4 Calib	ResNet50 calibration (pre/post temp scaling)	W3	M2
T29	5 XAI	ResNet50 explainability + faithfulness score	W3	M2
T34	6 Robust	ResNet50 external test + explanation-shift analysis	W3	M3
T45	9 Paper	Introduction + Related Work + Research Gap	W1-W4	M4
T49	9 Paper	Conclusion + Abstract	W4	M4
T51	9 Paper	Reference management (Zotero) — ongoing	W1-W4	M4

ALSO SUPPORTING (assist / review)

ID	Task / Sub-issue	Owner	Week
T04	Define shared	M5	W1
T05	Download COVID-19	M2	W1
T09	Shared PyTorch	M2	W1
T11	Train single baseline	M2	W1
T33	External-test protocol	M5	W3
T41	Consolidate all metrics	M5	W4
T42	Statistical comparison	M5	W4
T46	Methodology section	M2	W2-W4
T47	Experiments & Results	M5	W4
T50	Presentation slides	M3	W4

T52	Final proofread +	M5	W4

Depends on	Est. days	Deliverable
-	1	Repo + README + branch rules
T01	1	Tracking project + seed policy
T09,T20	1	ResNet50 frozen checkpoint
T12	2	ResNet50 final + curves
T23,T13	1	ResNet50 ECE + reliability
T28	1	ResNet50 heatmaps + score
T33,T13	1	ResNet50 OOD result
-	3	Paper sect. 1-2
T47	1	Paper sect. 6 + abstract
-	1	Bibliography

Member 2 — DenseNet121 & Data Pipeline Lead

TASKS YOU OWN (accountable)

ID	Epic	Task / Sub-issue	Week	M.stone
T05	1 Data	Download COVID-19 Radiography dataset; integrity check; class-count report	W1	M0
T06	1 Data	Build preprocessing module (resize 224, normalise, grayscale->3ch)	W1	M0
T07	1 Data	Stratified 70/15/15 split with fixed seed; save index files	W1	M0
T08	1 Data	Augmentation module (rotation, flip, jitter, zoom, shift, crop)	W1	M0
T09	1 Data	Shared PyTorch DataLoader with class weights for imbalance	W1	M0
T11	2 Baseline	**Train single baseline (DenseNet121) end-to-end to validate pipeline [GATE M0]**	W1	M0
T14	2 Train	DenseNet121 — Phase 1 frozen training	W2	M1
T15	2 Train	DenseNet121 — Phase 2 fine-tune + tuning	W2	M1
T25	4 Calib	DenseNet121 calibration	W3	M2
T30	5 XAI	DenseNet121 explainability + faithfulness	W3	M2
T35	6 Robust	DenseNet121 external test	W3	M3
T46	9 Paper	Methodology section	W2-W4	M4

ALSO SUPPORTING (assist / review)

ID	Task / Sub-issue	Owner	Week
T01	Create GitHub repo,	M1	W1
T03	Compute setup:	M3	W1
T04	Define shared	M5	W1
T10	Download external	M5	W1-W2
T20	Standardise shared	M5	W1-W2
T21	Train/adapt U-Net lung-	M5	W2
T22	Generate lung masks	M5	W2-W3
T23	Calibration standard	M5	W2-W3

T38	Full-image vs lung-	M5	W3
T47	Experiments & Results	M5	W4
T50	Presentation slides	M3	W4
T51	Reference management	M1	W1-W4
T52	Final proofread +	M5	W4

Depends on	Est. days	Deliverable
-	1	Verified dataset + EDA notes
T04	1	preprocess.py
T06	1	split index files
T06	1	augment.py
T07,T08	1	dataloader.py (shared)
T09	2	Working baseline + confusion matrix
T09,T20	1	DenseNet frozen checkpoint
T14	2	DenseNet final + curves
T23,T15	1	DenseNet ECE + reliability
T28	1	DenseNet heatmaps + score
T33,T15	1	DenseNet OOD result
-	2	Paper sect. 3

Member 3 — EfficientNet-B0 & Efficiency Lead

TASKS YOU OWN (accountable)

ID	Epic	Task / Sub-issue	Week	M.stone
T03	0 Setup	Compute setup: Colab/Kaggle GPU, requirements.txt, env pinning	W1	M0
T16	2 Train	EfficientNet-B0 — Phase 1 frozen training	W2	M1
T17	2 Train	EfficientNet-B0 — Phase 2 fine-tune + tuning	W2	M1
T26	4 Calib	EfficientNet-B0 calibration	W3	M2
T31	5 XAI	EfficientNet-B0 explainability + faithfulness	W3	M2
T36	6 Robust	EfficientNet-B0 external test	W3	M3
T39	7 Effic	**Efficiency harness: params, FLOPs, latency (CPU/GPU), GPU memory [GATE E8]**	W3	M3
T40	7 Effic	Run efficiency benchmark on all 4 models	W3	M3
T43	8 Analysis	Pareto trade-off analysis (accuracy vs trust vs efficiency)	W4	M4
T50	9 Paper	Presentation slides	W4	M4

ALSO SUPPORTING (assist / review)

ID	Task / Sub-issue	Owner	Week
T01	Create GitHub repo,	M1	W1
T04	Define shared	M5	W1
T06	Build preprocessin	M2	W1
T11	Train single baseline	M2	W1
T28	XAI standard:	M4	W2-W3
T44	Generate all figures	M4	W4
T47	Experiments & Results	M5	W4
T51	Reference management	M1	W1-W4
T52	Final proofread +	M5	W4

Depends on	Est. days	Deliverable
T01	1	Reproducible environment
T09,T20	1	EffNet frozen checkpoint
T16	2	EffNet final + curves
T23,T17	1	EffNet ECE + reliability
T28	1	EffNet heatmaps + score
T33,T17	1	EffNet OOD result
-	1	efficiency.py
T39,T13,T15,T17,T19	1	Efficiency table
T41	1	Pareto chart + ranking
T41	2	Slide deck

Member 4 — ViT-Base & Explainability Lead

TASKS YOU OWN (accountable)

ID	Epic	Task / Sub-issue	Week	M.stone
T18	2 Train	ViT-Base — Phase 1 frozen training	W2	M1
T19	2 Train	ViT-Base — Phase 2 fine-tune (DeiT/ViT-S fallback if unstable)	W2	M1
T27	4 Calib	ViT-Base calibration	W3	M2
T28	5 XAI	**XAI standard: Grad-CAM (CNN), Attention Rollout (ViT), SHAP + faithfulness/localisation score [GATE E6]**	W2-W3	M2
T32	5 XAI	ViT-Base attention rollout + faithfulness	W3	M2
T37	6 Robust	ViT-Base external test	W3	M3
T44	8 Analysis	Generate all figures (curves, heatmaps, reliability, Pareto)	W4	M4
T48	9 Paper	Discussion + model-selection guidance	W4	M4

ALSO SUPPORTING (assist / review)

ID	Task / Sub-issue	Owner	Week
T01	Create GitHub repo,	M1	W1
T04	Define shared	M5	W1
T08	Augmentation module	M2	W1
T11	Train single baseline	M2	W1
T29	ResNet50 explainability	M1	W3
T30	DenseNet121	M2	W3
T31	EfficientNet-B0	M3	W3
T47	Experiments & Results	M5	W4
T50	Presentation slides	M3	W4
T51	Reference management	M1	W1-W4
T52	Final proofread +	M5	W4

Depends on	Est. days	Deliverable
T09,T20	1	ViT frozen checkpoint
T18	2	ViT final + curves
T23,T19	1	ViT ECE + reliability
T22	2	xai.py + faithfulness metric
T28	1	ViT attention maps + score
T33,T19	1	ViT OOD result
T41	1	Figure pack
T41	2	Paper sect. 5

Member 5 — Segmentation/Eval & Paper Lead

TASKS YOU OWN (accountable)

ID	Epic	Task / Sub-issue	Week	M.stone
T04	0 Setup	Define shared config: image size, split ratios, metrics schema, checkpoint format	W1	M0
T10	1 Data	Download external RSNA set, DICOM->PNG convert, map labels	W1-W2	M1
T20	2 Train	Standardise shared training-loop template (frozen->fine-tune)	W1-W2	M0
T21	3 Segment	Train/adapt U-Net lung-segmentation model	W2	M1
T22	3 Segment	Generate lung masks for primary + external sets	W2-W3	M1
T23	4 Calib	**Calibration standard module: ECE, Brier, reliability, temperature scaling [GATE E5]**	W2-W3	M2
T33	6 Robust	External-test protocol standard (label mapping, OOD metrics)	W3	M3
T38	6 Robust	Full-image vs lung-segmented experiment (shortcut mitigation)	W3	M3
T41	8 Analysis	Consolidate all metrics into master results tables	W4	M4
T42	8 Analysis	Statistical comparison (significance, confidence intervals)	W4	M4
T47	9 Paper	Experiments & Results section	W4	M4
T52	9 Paper	**Final proofread + reproducibility rerun from clean repo [GATE M4]**	W4	M4

ALSO SUPPORTING (assist / review)

ID	Task / Sub-issue	Owner	Week
T01	Create GitHub repo,	M1	W1
T02	Set up experiment	M1	W1
T07	Stratified 70/15/15	M2	W1
T11	Train single baseline	M2	W1
T24	ResNet50 calibration	M1	W3
T25	DenseNet121 calibration	M2	W3
T26	EfficientNet-B0	M3	W3
T27	ViT-Base calibration	M4	W3

T34	ResNet50 external test	M1	W3
T35	DenseNet121 external	M2	W3
T36	EfficientNet-B0 external	M3	W3
T37	ViT-Base external test	M4	W3
T39	Efficiency harness:	M3	W3
T40	Run efficiency	M3	W3
T43	Pareto trade-off analysis	M3	W4
T45	Introduction + Related	M1	W1-W4
T48	Discussion + model-	M4	W4
T49	Conclusion + Abstract	M1	W4
T50	Presentation slides	M3	W4
T51	Reference management	M1	W1-W4

Depends on	Est. days	Deliverable
T01	1	config.yaml + metrics spec
T06	2	External test set ready
T09	1	train_template.py
T06	2	Segmentation model
T21	1	Mask sets
-	1	calibration.py (shared)
T10	1	external_eval.py
T22	2	Segmentation ablation result
T24,T29,T34,T40	1	Master results tables
T41	1	Stats appendix
T41	2	Paper sect. 4
T47,T48,T49	1	Camera-ready + repro log

STATUS DASHBOARD (auto from Master Task Li

Total tasks	52
Not started	52
In progress	0
Blocked	0
Done	0
% Complete	0.0%

Owned tasks per member

Member	Owned tasks	Done
M1	10	0
M2	12	0
M3	10	0
M4	8	0
M5	12	0